MBI 2026 · S24 · Nakamura — Anticipated Q&A

48 questions · English answers (30–60 s) with Japanese key points · priority: A Very likely B Likely C Possible / tough

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Useful phrases

Key numbers (from the paper)

Sample	SCZ 104 vs HC 101; 7 hospitals + 3 clinics; 2018–2024; age ≥ 20; DSM-5/ICD-10
Age / sex	48.3 ± 12.7 vs 51.3 ± 18.1 y ($p = 0.166$); female 48.1% vs 55.4% ($p = 0.360$)
PANSS (SCZ)	Total 63.3 ± 18.0; positive 15.4; negative 16.4; general 31.4 → mostly mild–moderate
Education	Paper: 13.3 ± 2.4 ($N = 51$) vs 15.4 ± 2.8 ($N = 70$), $p < 0.001$. ⚠ Slide 14 shows 13.1 ($N = 44$)
Visits / tasks	2–5 visits, ≥ 1 month apart; free conversation, picture description (VPTA, COGNISTAT, WAB), storytelling (Cinderella / Kaguya-hime); 30–60 min
Features	76 features: GiNZA (POS, NER, dependency), Word2Vec (WikiEntVec, Suzuki 2018) + TF-IDF, SentenceBERT
Split	Participant-level 80/20 train/test, stratified by diagnosis, sex, age group; missing values rare (0.22% Task 1)
Exploratory	Univariate GEE, FDR $q < 0.05$: 35/76 features significant in free conversation
Factor analysis	Horn's parallel analysis → 9 factors, Varimax; 62.5% variance (F1 19.6%); representative = loading ≥ 0.60 and ≥ 0.20 above next; all VIF < 3
Core markers (Task 1)	Mean Pairwise Word Similarity ↑ (OR ≈ 2.3 per SD); Case-particle ratio ↓; Adverb ratio ↓ (adjusted for age, sex)
Performance	Held-out AUC 0.87 (95% CI 0.74–0.97; permutation $p < 0.001$); balanced accuracy 0.83; sensitivity 0.76; specificity 0.90
Cross-task	Partial conjunction: Adverb ratio qPC = 0.021; Case particle ratio qPC = 0.014; Adverb × task interaction Wald $\chi^2(2) = 35.2$, $p < 0.001$; case particle $\beta = -0.66$, no interaction
Semantic network	SCZ: clustering ↑, density ↑; average distance ↓, diameter ↓, closeness centrality ↓; TF-IDF top 10% words, similarity threshold 0.3, NetworkX
Software	Stata 17 (GEE, bootstrap, permutation); Python (scikit-learn, statsmodels, scipy, pingouin)
Paper	Nakamura H, et al. Psychological Medicine 2026;56:e193. doi:10.1017/S0033291726104668

Top 5 — 30-second answers

Q. Is this specific to schizophrenia?
Not yet tested. Healthy-control comparison only; UNDERPIN also has depression, bipolar, anxiety and MCI/dementia, so transdiagnostic analysis is the next step.

Q. Does it generalise beyond Japanese?
The dimensions probably do; the surface markers depend on grammar. Fewer adverbs also in English (Tang 2021); Turkish parallels; semantic narrowing across languages.

Q. Medication effect?
Cannot exclude; dose not modelled. Markers track PANSS severity and match unmedicated/high-risk literature, but first-episode and longitudinal data are needed.

Q. AUC vs healthy controls — useful?
Diagnosis is not the goal. The value is turning vague impressions into reproducible numbers for monitoring change within a person.

Q. Why not LLMs?
Interpretability and sample size (~200). LLMs are powerful but opaque; we want markers a clinician can explain. LLM features are a future comparison.

↑ contents

A. Study design and methods

A1 Why did you use three different speech tasks? Wouldn't one standardized task be simpler?

A

Each task places different demands on the speaker. Free conversation is open and spontaneous, picture description is anchored to a visual stimulus, and storytelling requires narrative planning and memory. Using all three let us ask whether a marker reflects the illness itself or just the demands of one task. That is exactly what we found: case-particle reduction was task-invariant, while the adverb effect was strongest in free conversation. So multitask designs help separate stable, trait-like markers from context-dependent ones.

> 課題ごとに負荷が異なる。3課題で、課題に依存しない指標（格助詞）と課題依存的な指標（副詞）を区別できた。

Basis: Slide 11, 22 · ↑ contents

A2 How did you avoid overfitting with 76 features and only about 200 participants?

A

Three safeguards. First, we split the data at the participant level, 80% training and 20% test, so visits from the same person never appeared in both sets. Second, all preprocessing, standardization and feature selection were done only on the training set. Third, we did not let the classifier choose among 76 features; we used factor analysis, which is independent of diagnosis, to reduce them to nine representative features. Performance was then checked on the untouched test set, with bootstrap confidence intervals and a permutation test.

> 参加者単位で8:2に分割、前処理と特徴量選択は学習データのみ、診断と独立な因子分析で9個に削減、テストデータで評価、ブートストラップと並べ替え検定。

Basis: Slide 13 · ↑ contents

A3 Why did you choose factor analysis and one representative feature per factor, instead of LASSO or a machine-learning model? B

Our priority was interpretability. LASSO or random forests select features by their association with diagnosis, which can be unstable with correlated features and hard to interpret. Factor analysis groups correlated features into latent dimensions without using the diagnosis, and choosing one representative feature per factor keeps each predictor meaningful. It also kept collinearity very low; all variance inflation factors were below three. The trade-off is that we may lose some predictive power, but each marker can be explained to a clinician.

> 解釈性を優先。因子分析は診断を使わずに相関する特徴量をまとめるので、選択が安定し解釈しやすい。VIFはすべて3未満。

Basis: Slide 13, 21 · ↑ contents

A4 How did you decide on nine factors? B

We used Horn's parallel analysis, which compares the eigenvalues of our data with those from random data. It supported nine factors, and the solution remained stable across bootstrap replications. After Varimax rotation, the nine factors explained 62.5% of the variance; the first factor explained 19.6%, and the others contributed fairly evenly. For each factor, the representative feature needed an absolute loading of at least 0.60 and a margin of at least 0.20 over its next loading.

> Hornの平行分析で9因子、ブートストラップでも安定。Varimax回転後の累積寄与率は62.5%。代表特徴量は負荷量0.60以上かつ次点より0.20以上大きいもの。

Basis: Paper; Slide 21 · ↑ contents

A5 Why GEE rather than a simple logistic regression or a mixed model? B

Each participant contributed two to five visits, so observations are not independent. Generalized estimating equations with robust standard errors account for this clustering and give population-averaged estimates, which is what we want for group differences. Mixed models are an alternative, but GEE is robust to misspecification of the correlation structure, and with our sample it was the more conservative choice.

> 1人2〜5回の来院で観察が独立でないため。GEEは参加者内相関を考慮し、相関構造の誤指定に頑健。

Basis: Slide 13 · ↑ contents

A6 How did you handle differences in speech length? Patients may simply talk less. B

This is important, because many raw counts scale with the amount of speech. We expressed features as ratios where appropriate, for example adverbs per morpheme, and normalized by the number of morphemes when needed. We then z-standardized features using training-set parameters. The number of sentences itself was one of the representative features, and it was not significant in the final model, which suggests our core markers are not simply a by-product of talking less.

> 比率化と形態素数での正規化で発話量を補正。文の数自体も代表特徴量に含めたが最終モデルで有意でなく、主要3指標は発話量の副産物ではないと考えられる。

Basis: Slide 13, 21 · ↑ contents

A7 Speech was transcribed by humans. How reliable were the transcripts, and how did you treat fillers and disfluencies? B

Transcription was done verbatim by a professional service, without normalization or removal of disfluencies, and only the participant's speech was analyzed. Manually inserted punctuation defined sentence boundaries for sentence-level measures, while dependency parsing was run after removing punctuation so that GiNZA determined boundaries automatically. Keeping disfluencies matters, because fillers and fragments are part of what we want to measure.

> 専門業者が逐語的に書き起こし、言い淀みも保持。文単位の指標は人手の句読点、依存構造解析は句読点を除いてGiNZAが自動判定。

Basis: Paper methods · ↑ contents

A8 Which Word2Vec model did you use, and does the result depend on it? C

We used a publicly available Japanese Word2Vec model trained on Wikipedia, WikiEntVec. We chose a static, transparent model on purpose, so that similarity values are easy to interpret. But it is a limitation: the semantic measures may depend on the embedding model and its training corpus. Replication with other embeddings, such as fastText or transformer-based models, is needed, and this is on our list.

> Wikipediaで学習した公開の日本語Word2Vec（WikiEntVec）。解釈しやすさのため静的モデルを選んだが、モデル依存性は限界で、他の埋め込みでの再検証が必要。

Basis: Slide 12; paper limitations · ↑ contents

A9 The confidence interval of the AUC is wide, 0.74 to 0.97. How confident are you in 0.87? C

That is fair. The test set was about 40 participants, so the interval is wide. What gives me confidence is that the permutation test was clearly significant, the balanced accuracy was 0.83, and the same markers appeared consistently across tasks. But the exact number should be confirmed in an external sample, ideally from different hospitals and languages.

> テストデータは約40名のため信頼区間が広い。並べ替え検定は有意、バランス精度0.83、課題横断で一貫。正確な値は外部検証で確認が必要。

Basis: Slide 21 · ↑ contents

B. Interpreting the results

B1 Case particles are often dropped in casual Japanese. Isn't reduced particle use just a normal conversational style? A

Particle omission does occur in everyday Japanese, especially in casual speech. But controls were interviewed in the same setting and style, so the difference is beyond normal omission. Case particles mark who did what to whom, so their reduction suggests less explicit marking of relations between words. We interpret it as reduced deployment of explicit relational marking under the demands of spontaneous production, such as planning and monitoring. Importantly, the reduction was found in every task, not only in casual conversation.

> 日常会話でも助詞の省略は起こるが、同じ条件の健常者との差であり通常の省略を超える。全課題で減少しており、会話スタイルだけでは説明できない。

Basis: Slide 21, 22, 23 · ↑ contents

B2 Why adverbs? What does reduced adverb use mean clinically? A

In Japanese, adverbs modify not only verbs but also adjectives and whole sentences, adding manner, degree, time and the speaker's stance. Among the major parts of speech, only adverbs were reduced, not nouns, verbs or adjectives. The adverb factor also loaded on discourse-level features. So we interpret it not as a simple grammatical deficit, but as weakened contextual modulation: a reduced capacity to elaborate, nuance and frame content. Clinically, this resembles the impoverished, flat quality of speech described in classical psychopathology.

> 日本語の副詞は文全体を修飾し、様態・程度・話者の態度を加える。主要品詞で減ったのは副詞のみ。単なる文法障害ではなく、文脈による調整機能の低下と解釈。臨床的には平板で貧困な発話に対応。

Basis: Slide 21, 23 · ↑ contents

B3 Higher word similarity sounds like more coherent speech. Why is it a sign of illness? A

Good question. Mean pairwise word similarity measures how close the chosen words are to each other in semantic space. Higher values mean the speaker stays within a narrow semantic neighborhood. Healthy speech moves flexibly between related and more distant concepts. So higher similarity reflects semantic narrowing, a restricted exploration of meaning, rather than better coherence. It fits with reduced informational density described in earlier work, and with our network finding: locally dense but globally poorly integrated.

> 単語間類似度が高い＝意味的に近い語ばかり使う＝意味探索の狭小化。一貫性が高いという意味ではない。ネットワーク所見（局所的に密、大域的に統合不良）と整合。

Basis: Slide 19, 21, 23 · ↑ contents

B4 Your network is denser and has shorter distances, yet you say information transfer is less efficient. Isn't that contradictory? B

It sounds contradictory, so let me be precise. Shorter average distance and smaller diameter mean the network is compact: the words a patient uses are all close together. But this compactness comes from staying in one region of semantic space, not from reaching many regions efficiently. The lower closeness centrality suggests that individual words are less well connected to the rest of the network. So the space explored is small and tightly clustered, and distant ideas are rarely reached. I agree the phrase 'information transfer' is a metaphor, and it should be tested more formally.

> 距離が短い＝狭い領域にまとまっているためで、広い意味空間を効率よく移動できているわけではない。中心性の低下は各語の結びつきの弱さを示す。「情報伝達効率」は比喩であり、厳密な検証が必要。

Basis: Slide 19 · ↑ contents

B5 Your hypothesis was reduced syntactic complexity. Was it supported? B

Partly. In free conversation, classic complexity measures such as normalized tree height were not significant, perhaps because unconstrained conversation lets speakers avoid complex structures. In storytelling, normalized dependency distance was reduced, which fits the working-memory demands of narrative. And the strongest syntax-related finding was the reduction of case particles, which we see as reduced explicitness rather than reduced complexity. So the hypothesis was refined: the syntactic change is better described as simplification of relational marking.

> 一部支持。自由会話では木の高さなどは有意でなかったが、物語課題で依存距離が低下。最も強い所見は格助詞の減少で、「複雑性の低下」より「関係標識の明示性の低下」と言い換えるのが適切。

Basis: Slide 7, 22; paper · ↑ contents

B6 Patients used more person-related named entities. How do you interpret that? B

This was an exploratory finding. It is broadly consistent with earlier reports of altered referential structure and increased noun-phrase density in psychotic discourse. One possibility is that patients refer to people more explicitly, or talk more about other people, which could relate to social or persecutory themes. But it did not remain a representative feature in the final model, so I would not over-interpret it.

> 探索的所見。精神病の談話における参照構造の変化や名詞句密度の増加の報告と整合。被害的・対人的な話題との関連も考えられるが、最終モデルには残っていないので過大解釈は避ける。

Basis: Slide 16 · ↑ contents

B7 Perseveration was strongly elevated only in storytelling. Why? B

Storytelling requires sequencing events and moving the narrative forward, so repeating the same expression stands out. In free conversation, topics shift naturally, and in picture description the stimulus constrains content. So perseveration may be most visible when a task requires sustained narrative progression. It mirrors the clinical phenomenon of perseveration, and it suggests task design matters for which symptoms we capture.

> 物語課題は出来事を順に進める必要があり、同じ表現の繰り返しが目立つ。課題設計によって捉えられる症状が変わることを示す。

Basis: Slide 22 · ↑ contents

B8 How do your markers relate to symptoms? Did you correlate them with PANSS? B

Yes. The core markers were associated with PANSS severity, and the clustering coefficient correlated with PANSS negative symptoms and with the PANSS item on fluency of speech. So our profile seems closer to the negative, impoverished side of thought disorder than to florid derailment. But we did not use a dedicated thought-disorder scale such as the TLC, so a direct mapping to thought-disorder phenomenology is still needed.

> 主要指標はPANSS重症度と関連、クラスター係数は陰性症状と発話の流暢性項目に相関。陰性・貧困型の思考障害に近いが、TLC等での直接検証は未。

Basis: Slide 17, 23 · ↑ contents

B9 How large are the effects? Could a clinician notice these differences in a conversation? C

For word similarity, one standard deviation was associated with roughly 2.3-fold higher odds of schizophrenia, and the case-particle and adverb effects were of a similar or larger magnitude in the opposite direction. These are group-level differences, and they are subtle at the level of a single sentence. A clinician would probably sense them only as an overall impression, such as flat or impoverished speech. That is exactly why quantification is useful: it makes these subtle patterns measurable and comparable over time.

> 単語類似度は1SDでオッズ約2.3倍。格助詞・副詞も同程度以上（逆方向）。1文単位では気づきにくく、臨床では「平板」などの印象として感じる程度。だからこそ定量化が役立つ。

Basis: Slide 21 · ↑ contents

B10 What is new compared with previous NLP studies of schizophrenia? C

Three things. First, scale and breadth: over 200 participants, three tasks and 76 features across four linguistic levels, analysed together rather than one at a time. Second, an interpretable reduction of these features into a three-part phenotype, tested for cross-task stability. Third, language: this is one of the largest studies in Japanese, an agglutinative language, which reveals markers such as case particles that are hard to see in English.

> 規模と幅（200名超・3課題・76特徴量・4レベル統合）、解釈可能な三側面の表現型と課題横断性の検証、日本語（膠着語）での大規模研究。

Basis: Slide 7, 8, 23 · ↑ contents

C. Confounders and limitations

C1 How do you rule out the effects of antipsychotic medication? A

We cannot fully rule them out. We did not model dose, and sedation or extrapyramidal effects could reduce speech elaboration. Three points, however. The markers were related to PANSS severity, not only to group membership. Semantic narrowing has been reported in unmedicated and high-risk samples in other studies. And case particles are a grammatical choice rather than a motor feature. Still, first-episode, drug-naive and longitudinal data with dose information are the proper test.

> 完全には除外できない（用量は未調整）。ただし指標はPANSS重症度と関連し、未服薬・高リスク群の先行研究でも意味的狭小化が報告され、格助詞は運動症状ではなく文法的選択。初発・未服薬・縦断データでの検証が必要。

Basis: Limitations · ↑ contents

C2 Education differed between groups. Could that explain the findings? A

It is a fair concern. Education data were available only for a subset, so we did not include it in the primary models. In a complete-case sensitivity analysis adjusting for education, all three core markers remained significant. Future studies should also measure premorbid IQ and verbal ability directly.

> 教育歴は一部のみ取得。教育歴で調整した感度分析でも主要3指標は有意のまま。病前IQや言語能力の直接測定が今後の課題。

Basis: Slide 14 (⚠ slide shows N = 44; paper N = 51) · ↑ contents

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C3 Could cognitive impairment, rather than schizophrenia itself, explain the language changes? B

Very possibly in part, and that is not necessarily a confound. Language production depends on working memory, planning and semantic retrieval, which are often impaired in schizophrenia. We did not have direct cognitive measures in this analysis, so we cannot separate the two. I see this as an important next step: linking linguistic markers to cognition and social functioning, because that would also make them more clinically meaningful.

> 一部は認知機能障害を反映している可能性が高いが、必ずしも交絡ではない。本解析では認知機能の直接測定がなく分離できない。認知・社会機能との関連付けが次の重要課題。

Basis: Slide 23 · ↑ contents

C4 Your patients were chronic and middle-aged, around 48 years old. Do the findings apply to early psychosis? B

That is an important limitation. Our sample reflects established, mostly mild-to-moderate illness, with a mean age around 48 years. So we cannot say whether these markers are present at onset or develop over the course of illness. Testing them in first-episode and clinical high-risk cohorts is essential, both scientifically and clinically, because that is where early detection matters most.

> 対象は平均48歳前後の慢性期・軽症〜中等症。発症時から存在するか経過で生じるかは不明。初発・高リスク群での検証が不可欠。

Basis: Slide 14, 24 · ↑ contents

C5 Free conversation depends on the interviewer and the topic. How did you control for that? B

Interviewers followed a semi-structured guide, kept a neutral style with minimal prompting, and wore headset microphones so we could separate speakers. We analyzed only the participant's speech, and repeated visits were modeled with GEE, which reduces the influence of any single session. Still, interviewer effects are real, and standardized prompts, or brief self-recorded speech at home, would improve comparability.

> 半構造化・中立的な面接、ヘッドセットで話者分離、参加者の発話のみ解析、複数回来院をGEEで考慮。面接者効果は残るので標準化プロンプトや自宅録音が改善策。

Basis: Slide 11 · ↑ contents

C6 Could regional dialects in Japan affect particle or adverb use? C

Yes, dialects can affect particle forms and some adverbs. Our participants came from multiple sites, and both groups were recruited in similar regions, which reduces systematic bias. But we did not model dialect explicitly. This is a good example of the bias concern I raised on the ethics slide: models must be checked across dialects and backgrounds before clinical use.

> 方言は助詞や副詞に影響しうる。両群とも同様の地域から多施設で募集したが、方言は明示的に調整していない。倫理スライドで述べたバイアスの好例で、臨床応用前に検証が必要。

Basis: Slide 27 · ↑ contents

C7 Why not use written language or acoustic features as well? C

We focused on transcribed spoken language to isolate linguistic content. Written language may capture semantic coherence differently, and acoustic features such as pauses, prosody and speech rate carry complementary information, especially for negative symptoms. Combining text, voice and other signals, such as the wearable data Dr. Liang presented, is a natural next step.

> 言語内容を分離するため書き起こした話し言葉に限定。書き言葉や音響特徴（間・韻律・話速）は補完的な情報を持つ。Liang先生のウェアラブルを含むマルチモーダル統合が次の段階。

Basis: Slide 24 · ↑ contents

D. Generalizability

D1 Do these findings generalize to other languages, such as English or Chinese? A

Partly, I think. Semantic narrowing has been reported across several languages, and Tang and colleagues found fewer adverbs in English-speaking patients as well. Case particles are specific to Japanese, but the underlying change, less overt marking of who did what to whom, has parallels in Turkish, another agglutinative language, where reduced determiners and referential instability have been reported. So I expect the dimensions to generalize while the surface markers depend on grammar. Mandarin has few inflections and relies more on word order and particles, so it would be a very interesting test. I would be happy to collaborate.

> 意味的狭小化は多言語で報告、副詞減少は英語でも（Tang 2021）。格助詞は日本語固有だが、関係標識の低下はトルコ語に類似所見。次元は一般化し、表層の指標は文法に依存すると予想。中国語での検証は非常に興味深く、共同研究を歓迎。

Basis: Slide 23, 26 · ↑ contents

D2 Are these markers specific to schizophrenia, or do they also appear in depression or bipolar disorder? A

That is the key question, and this study cannot answer it, because we compared patients only with healthy controls. Reduced adverbs and less elaboration could also reflect psychomotor slowing or negative symptoms seen in depression. UNDERPIN also includes depression, bipolar disorder, anxiety disorders and mild cognitive impairment or dementia, so transdiagnostic comparison is our next analysis. I suspect some markers are shared, and some, perhaps semantic network structure, are more specific.

> 健常者との比較のみで特異性は未検証。副詞減少はうつ病の精神運動抑制でも起こりうる。UNDERPINには他疾患があるので疾患横断比較が次の解析。共通の指標と特異的な指標があると予想。

Basis: Slide 24 · ↑ contents

D3 Japanese has no articles and allows omission of subjects. Does that make Japanese better or worse for detecting these changes? B

In some ways better. Japanese is agglutinative, so grammatical relations are marked by separate, segmentable morphemes such as case particles. That makes reductions in relational marking easy to count. Adverbs in Japanese also have a very broad modifying role, so reduced modification may be especially visible. On the other hand, subject omission is normal in Japanese, so some referential measures used in English are less informative. Typology shapes which disturbances are most detectable.

> 膠着語で文法関係が分節可能な形態素（格助詞）で示されるので、関係標識の減少を数えやすい。副詞の修飾範囲も広い。一方、主語省略は通常なので英語の参照指標は使いにくい。類型論が検出しやすさを左右する。

Basis: Slide 23 · ↑ contents

E. Clinical application

E1 An AUC of 0.87 against healthy controls: is that clinically useful? Psychiatrists can already tell the difference. A

I agree that distinguishing patients from healthy controls is not the clinical goal. The value lies elsewhere. These markers turn vague impressions, such as impoverished speech, into reproducible numbers along three dimensions. That makes them candidates for monitoring severity, relapse and treatment response within the same person, where small changes are hard to see by eye. The diagnostic accuracy is mainly a check that the markers carry real signal.

> 健常者との鑑別自体は目的ではない。曖昧な臨床印象を再現可能な数値にし、同一患者の重症度・再発・治療反応の変化を追えることに価値がある。AUCは信号があることの確認。

Basis: Slide 27 · ↑ contents

E2 What is your next step toward the clinic?

A

Longitudinal monitoring. We want to see whether the same interpretable markers track symptom change, relapse and treatment response within individuals, ideally from brief speech tasks that can be recorded remotely. In parallel, we plan transdiagnostic comparisons within UNDERPIN, cross-linguistic replication, and integration with acoustic and wearable data.

> 縦断モニタリング（同一患者内で症状変化・再発・治療反応を追う、短い遠隔録音が理想）。並行して疾患横断比較、他言語での再現、音声・ウェアラブルとの統合。

Basis: Slide 26, 28 · ↑ contents

E3 You used manual transcription. Can this work with automatic speech recognition in real clinics?

B

Manual transcription was deliberate, to separate language effects from recognition errors. Japanese speech recognition has improved greatly, and modern systems are good enough for many purposes. But they tend to smooth over disfluencies, fillers and fragments, which are part of what we measure, and errors may differ between patients and controls. So before clinical use, each feature must be validated on recognition output against manual transcripts.

> 認識誤りと言語の変化を分けるため人手で書き起こし。日本語の自動音声認識は大きく向上したが、フィラーや言い淀みを消しがちで、群間で誤りの傾向が異なる可能性もある。臨床応用前に特徴量ごとの検証が必要。

Basis: Slide 12 · ↑ contents

E4 How much speech do you need for a reliable measurement?

B

In this study, interviews lasted 30 to 60 minutes, with picture description and storytelling together taking about ten minutes. We have not yet formally determined the minimum length for reliable estimates. That is an important practical question for monitoring, and we plan to examine how stable each feature is with shorter samples, for example a few minutes of speech.

> 本研究は30〜60分（絵の描写と物語で約10分）。信頼できる最小発話量はまだ未検討で、短い発話での各特徴量の安定性を調べる予定。

Basis: Slide 11 · ↑ contents

E5 Could this be used for early detection or screening in the general population? B

Possibly in the future, but we should be careful. Our data come from established illness, so we cannot claim early detection. Screening in low-prevalence populations also produces many false positives, which can cause anxiety and stigma. I think the most realistic first use is monitoring people already in care, with their consent, rather than screening the general population.

> 将来的には可能性があるが慎重であるべき。本研究は発症後のデータで早期発見は主張できない。有病率の低い集団でのスクリーニングは偽陽性が多くスティグマにもつながる。まずは同意を得た治療中患者のモニタリングが現実的。

Basis: Slide 24, 27 · ↑ contents

E6 How would you integrate this with Dr. Liang's wearable markers? C

They are complementary. Wearables capture physiology and behavior, such as sleep, activity and heart-rate variability, continuously and passively. Language captures the content and organization of thought. Combining them could tell us not only that something has changed, but what kind of change it is, for example negative symptoms versus mood. A multimodal, longitudinal design within the same patients would be a very natural collaboration.

> 相補的。ウェアラブルは生理・行動を受動的・連続的に、言語は思考の内容と構造を捉える。統合すれば「変化があった」だけでなく「どのような変化か」まで分かる可能性。同一患者での縦断マルチモーダル研究は自然な共同研究。

Basis: Session context · ↑ contents

F. Large language models and generative AI

F1 Why did you use Word2Vec and handcrafted features instead of BERT or large language models? A

Two reasons. First, interpretability: each of our features maps onto a linguistic or clinical concept, so a clinician can understand what changed. Second, with about 200 participants, large end-to-end models risk overfitting. We did use SentenceBERT for sentence-level coherence. Comparing our interpretable markers with LLM-based features, and seeing what LLMs add, is a natural next step.

> 解釈性と、約200名では大規模モデルが過学習しやすいこと。文レベルではSentenceBERTを使用。LLM特徴量との比較は今後の課題。

Basis: Slide 5 · ↑ contents

F2 Could a large language model simply rate thought disorder from transcripts? B

Recent studies suggest that LLMs can predict expert ratings of coherence and tangentiality without task-specific training, and can detect loose associations. That is promising, especially for scaling up assessment. But LLM ratings need validation against clinicians, across languages and dialects, and their reasoning is hard to inspect. I see LLM ratings and interpretable markers as complementary: one for flexibility, the other for explanation.

> 近年、LLMが追加学習なしで専門家の評価（一貫性・脱線）を予測し、連合弛緩を検出できるとの報告がある。有望だが、臨床医との照合、多言語・方言での検証が必要で、推論過程は見えにくい。解釈可能な指標とは相補的。

Basis: Slide 26 · ↑ contents

F3 Could chatbots and generative AI be harmful for people with psychosis? A

Yes, that is a real concern. Østergaard warned in 2023 that chatbots might reinforce delusions in people prone to psychosis, and case reports have since appeared. Language models tend to agree with the user and to sound convincing, which is the opposite of what someone with a delusion needs. This is why I argue for clinician oversight, safety testing with people with lived experience, and clear limits on autonomous use in mental health.

> 現実的な懸念。Østergaard（2023）が妄想を強化しうると警告し、その後症例報告も出ている。LLMは利用者に同調し説得力がある。臨床医の監督、当事者を含めた安全性評価、自律的利用の明確な制限が必要。

Basis: Slide 27 · ↑ contents

F4 If large language models become very good, will psychiatrists still be needed for assessment? C

Yes. Assessment in psychiatry is not only classification; it is part of a relationship, and it carries responsibility for decisions that affect a person's life. AI can make our observations more precise and consistent, and free time for patients. But diagnosis and treatment decisions should remain in human hands, and the therapeutic relationship is itself part of care.

> 必要。精神科の評価は分類だけでなく関係性の一部であり、人生に関わる決定への責任を伴う。AIは観察を精密にし時間を生むが、診断と治療の決定は人間に残すべき。

Basis: Slide 28 · ↑ contents

G. Ethics, privacy and governance

G1 How would you protect privacy if speech is monitored continuously? A

Three principles. Process as much as possible on the device, and store derived features rather than raw audio or text. Make consent specific, ongoing and easy to withdraw. And give patients access to their own data and a say in who sees the results. Language data are intimate data, so trust is a precondition for any benefit.

> 端末内処理で生データでなく特徴量のみ保存、同意は具体的・継続的・撤回可能に、患者が自分のデータにアクセスし共有範囲を決められること。信頼が利益の前提。

Basis: Slide 27, 28 · ↑ contents

G2 Can you really de-identify speech data? B

Not completely. Voice is biometric, and the content of speech often contains names, places and life events. Removing names helps, but people can sometimes be re-identified from content. That is why I prefer storing derived features, such as ratios and network measures, which are much harder to link back to a person, and keeping raw data under strict access controls.

> 完全には困難。声は生体情報で、内容にも固有名詞や出来事が含まれる。名前の削除だけでは再識別されうるので、比率やネットワーク指標などの派生特徴量を保存し、生データは厳格に管理する。

Basis: Slide 27, 28 · ↑ contents

G3 How do you address bias across cultures and languages? B

First, by testing models in diverse samples before use: different languages, dialects, education levels and cultures. Second, by reporting performance separately for subgroups, not only overall. Third, by preferring interpretable features, because when a marker behaves differently in a subgroup, we can see why. Our own Japanese results are a reminder that grammar shapes the markers, so a model trained in one language cannot be transferred blindly.

> 使用前に多様な集団（言語・方言・教育・文化）で検証、サブグループ別に性能を報告、解釈可能な特徴量を優先。日本語の結果自体が、文法が指標を左右し他言語へそのまま移植できないことを示している。

Basis: Slide 27 · ↑ contents

G4 Who should be responsible if an AI-based assessment is wrong? C

Responsibility should stay with clinicians and institutions, and systems should be designed so that high-stakes decisions, such as diagnosis, remain in human hands. AI outputs should be presented as information to support judgment, with their uncertainty visible. Developers also have responsibilities for validation, transparency and post-deployment monitoring, in line with medical-device and AI regulations.

> 責任は臨床医と医療機関に残し、診断など重大な決定は人間が行う設計に。AIの出力は不確実性を示した判断材料として提示。開発者も検証・透明性・導入後監視の責任を負う。

Basis: Slide 28 · ↑ contents

G5 Wouldn't continuous monitoring damage the therapeutic relationship? C

It could, if it is imposed or used to check up on patients. It could also strengthen it, if patients choose it, see their own data and use it together with the clinician to understand their illness. The difference lies in consent, transparency and purpose. That is why I emphasized trust as one of the three pillars.

> 押し付けや監視目的なら損なう。患者が選び、自分のデータを見て臨床医と一緒に病状理解に使うなら強化しうる。違いは同意・透明性・目的にあり、だから信頼を3本柱の一つにした。

Basis: Slide 27, 28 · ↑ contents

H. Mechanisms and neuroscience

H1 What is the neural basis of the locally dense, globally poorly integrated semantic network? B

We can only speculate. Our pattern parallels imaging findings of altered network organization in schizophrenia, for example more homogeneous within-module connectivity and reduced long-range integration. A PNAS study by Nour and colleagues in 2023 linked how patients move through semantic space during verbal fluency to hippocampal ripple activity measured with MEG. I work on ripples with intracranial EEG in epilepsy, so connecting semantic trajectories to hippocampal replay is a direction I personally find very exciting.

> 推測の域。統合失調症の脳ネットワーク（モジュール内の均質な結合、長距離統合の低下）と並行。Nour 2023（PNAS）は意味空間の移動と海馬リップルを関連付けた。自分の頭蓋内脳波リップル研究ともつながる。

Basis: Slide 23, 26 · ↑ contents

H2 Is semantic narrowing a cause of thought disorder or a consequence of something else, such as reduced motivation? C

We cannot tell from cross-sectional data. Semantic narrowing could reflect a primary change in semantic memory or retrieval, or it could follow from reduced motivation and negative symptoms, since the clustering coefficient correlated with negative symptoms. Longitudinal data, and links to cognitive and neural measures, would help to separate these explanations.

> 横断研究では判断できない。意味記憶・検索の一次的変化か、陰性症状（意欲低下）の結果か。クラスター係数が陰性症状と相関していることからも両方の可能性。縦断・認知・神経指標が必要。

Basis: Slide 17, 23 · ↑ contents

I. Data, reproducibility and collaboration

I1 Is your code or data available? B

The analysis code is available from the corresponding authors on reasonable request, as stated in the paper. Raw speech data cannot be shared openly because of privacy and consent, but we are open to collaboration under appropriate agreements, and harmonized cross-linguistic efforts such as the DISCOURSE consortium show how this can be done.

> 解析コードは論文に記載のとおり責任著者に請求可能。発話の生データはプライバシーと同意の関係で公開できないが、適切な契約の下での共同研究は歓迎。DISCOURSEのような多言語統合の枠組みが参考になる。

Basis: Paper data availability · ↑ contents

I2 Did you use large language models in writing the paper or this presentation? C

Yes, in a limited way. As stated in the paper, LLMs such as ChatGPT were used only to help refine English expression and the organization of the manuscript. They were not used for data analysis, scientific reasoning or generating hypotheses, and no LLM output was used without expert review. All analyses, interpretations and conclusions are our own.

> 限定的に使用。論文に明記のとおり、英文表現と構成の推敲のみに使用し、データ解析・科学的推論・仮説生成には使っていない。出力はすべて専門家が確認。

Basis: Paper acknowledgments · ↑ contents

Useful phrases

Buying time	That's an excellent question. Let me think for a moment.
Clarifying	If I understand correctly, you are asking whether … Is that right?
Didn't catch it	I'm sorry, could you repeat the question, a little more slowly?
No data yet	Honestly, we don't have data on that yet. My guess would be …, but it needs to be tested.

Out of scope	That goes beyond this study, but it's an important point, and I'd be glad to discuss it after the session.
Agreeing with a critique	I agree, that's a limitation of our study, and it's exactly what we want to address next.
Hand to the chair	Perhaps Professor Kishimoto would like to add something from the UNDERPIN perspective.
Linking to Dr. Liang	That connects nicely with Dr. Liang's talk; combining language and wearable data could …